Siddhi Kale

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https://github.com/siddhi-works

Summary

AI/ML Engineer with a B.Tech in AI and Data Science, specializing in multi-agent AI system design and data pipeline development. Proven ability to prototype intelligent product features and integrate AI tools for comprehensive verification. Delivered AI solutions for complex challenges.

Experience

Content Editor & Management Coordinator

Apr 2025 - Nov 2025

Abhivyakti Editorial Team, ADYPU

Pune, Maharashtra, India

- Managed content calendars across 5 social media platforms, implementing workflows that boosted production efficiency by 30% for timely delivery.
- Coordinated with cross-functional teams for 50+ content pieces quarterly, ensuring consistent output and alignment with strategic goals.
- Executed rigorous editing and proofreading protocols, maintaining brand consistency and reducing content errors by 15%, enhancing quality control.

Content & Marketing Team Volunteer

Nov 2021 - Mar 2023

Bindu Non-Profit Organisation

Remote

- Led a team of 5 content creators, increasing content output by 25% through strategic planning and effective coordination.
- Developed and executed a comprehensive content strategy that drove a 15% increase in organic traffic year-over-year.
- Oversaw marketing collateral creation, aligning content with strategic objectives and supporting lead generation efforts.

Research Volunteer

Sep 2024 - Present

NSS, ADYPU

Pune, Maharashtra, India

- Supported the planning and execution of 8 youth programs, impacting over 100 children annually with structured activities.
- Assisted in preparing program materials and resources, streamlining delivery processes for effective educational sessions.
- Managed participant registration and communication, maintaining accurate records for all attendees, demonstrating data organization skills.

Education

Bachelor of Technology in Artificial Intelligence and Data Science

Jul 2023 - Jul 2027

Ajeenkya Dy Patil University

Pune, India

AJEENKYA
D Y PATIL UNIVERSITY

Projects

ARIA : AI-Powered Research & Intelligence Agent

May 2026 - Jun 2026

- Designed a multi-agent workflow using LangGraph and LangChain to orchestrate reasoning, tool execution, and conversational memory for research automation.
- Integrated the arXiv API and PDF analysis pipelines to retrieve, extract, and synthesize insights from papers for literature reviews and idea generation.
- Automated publication-ready paper creation by generating structured LaTeX documents and exporting them as professionally formatted PDF reports.
- Developed a responsive Streamlit web application with real-time AI responses, persistent chat history, and downloadable research reports for users.

Python | Langgraph | Langchain | Google Gemini 2.5 Flash API | LaTeX | arXiv API | Git & GitHub | PyPDF2 | Streamlit

ClaimMate — AI-Powered Insurance Claim Companion

- Designed a supervisor-based multi-agent workflow with specialized Policy, Claim, Document, and Audit agents for comprehensive claim analysis.
- Built a context-aware claim reasoning pipeline, combining user inputs and agent outputs to maintain context across follow-up conversations.
- Implemented document and claim analysis to identify missing information, pending actions, and appropriate escalation paths for users.
- Developed and deployed a bilingual React + TypeScript web application with a Node.js/Express backend, integrating Gemini 2.5 Flash via secure APIs.

Generative AI | LLM | Agentic AI | GitHub | Render | React | Node.js | TypeScript | RESTful APIs

Resolve IQ : AI-Powered Intelligent Ticket Routing & Resolution Agent

- Implemented text preprocessing, TF-IDF feature extraction, and Logistic Regression for automated ticket classification across support categories.
- Developed a rule-based intent detection layer with ML fallback, enhancing handling of common, high-confidence issues.
- Built a knowledge retrieval module using historical ticket data to identify similar incidents and surface previously resolved solutions for users.
- Implemented confidence-based decision logic, priority estimation, and human-in-the-loop escalation for sensitive tickets, ensuring quality control.

Python | Scikit-learn | Logistic Regression | Pandas | NumPy | Flask | RESTful API | SQLite

AI-Powered Dermoscopic Skin Lesion Classification and Detection

Mar 2025 - Mar 2025

- Developed and implemented a deep learning model for classifying dermoscopic skin lesions, achieving 95% accuracy.
- Engineered image preprocessing pipelines to enhance lesion feature extraction and reduce noise for improved model performance.
- Integrated the model into a web application, providing real-time diagnostic assistance and showcasing product prototyping skills.
- Validated model performance against expert dermatologist diagnoses, demonstrating high concordance and reliability.

Python | TensorFlow | Keras | Scikit-learn | Flask | Pycharm

Projects

AI-Based Chest X-Ray Disease Detection Using ResNet-50

Apr 2025 - Apr 2025

- Developed and trained a ResNet-50 model for detecting diseases from chest X-rays, achieving an accuracy of 92% on a diverse dataset.
- Implemented data augmentation techniques to enhance model robustness and generalization, critical for medical image analysis.
- Preprocessed and curated a large-scale dataset of over 10,000 chest X-ray images, ensuring data quality.
- Fine-tuned pre-trained ResNet-50 layers to optimize performance for medical imaging, demonstrating transfer learning expertise.

Python TensorFlow Keras OpenCV Pandas NumPy Scikit-learn Matplotlib

ML-Based Intelligent Bot Detection Model for Social Media

Jan 2026 - Apr 2026

- Developed an R-based model to identify and mitigate bot activity on social media platforms, enhancing data integrity.
- Engineered feature extraction pipelines using R packages to process large-scale user interaction data effectively.
- Validated model performance, achieving 95% accuracy in detecting sophisticated bot networks for robust security.
- Reduced false positive rates by 60% through iterative refinement and hyperparameter tuning, optimizing detection accuracy.

R dplyr caret Shiny ggplot2 tidyr data.table

Plagiarism Detection System Using NLP

Nov 2025 - Nov 2025

- Developed a plagiarism detection system using Python to analyze text similarity across documents.
- Implemented natural language processing techniques, including TF-IDF and Cosine Similarity, for effective document comparison.
- Achieved a 90% accuracy rate in identifying duplicated content across a dataset of 1000 documents.
- Streamlined the text preprocessing pipeline for faster analysis and real-time plagiarism checks.

Python NLTK Scikit-learn Pandas Flask Streamlit

Technology

Programming & Web Development

Python SQL R JavaScript TypeScript Node.js React Flask Streamlit RESTful APIs Web Automation

AI & Machine Learning

Machine Learning Deep Learning Natural Language Processing Generative AI Large Language Models Agentic AI Transfer Learning Prompt Engineering TensorFlow Keras Scikit-learn LangChain LangGraph

Data Management & Tools

Data Analysis Data Visualization Pandas NumPy SQLite GitHub Tidyverse Text2Vec OpenCV PyPDF2 LaTeX Pycharm

Technology

Product & Workflow Skills

Workflow Design | Quality Control | Documentation | Problem Solving | Analytical Thinking | Attention to Detail | Teamwork | Communication | Content Strategy

Certificates

Tata - GenAI Powered Data Analytics Job Simulation
Forage

AWS re/Start Graduate
Amazon Web Services (AWS)

Artificial Intelligence and Machine Learning
Edunet Foundation

Data Science
CodeWithHarry

Achievements

Published Research Paper – ICAIML-AE 2026
Published "Hybrid Feature-Based Comparative Analysis of Supervised and Unsupervised ML-Based Intelligent Bot Detection Model for Social Media Data" in the International Conference on Artificial Intelligence and Machine Learning for Agro-Ecosystems (ICAIML-AE 2026), published by Atlantis Press (Springer Nature).

Languages

English ●●●●●

Hindi ●●●●○

Marathi ●●●○○